

## P3 — Effect modification

Week 5 · applies L4 (Interaction) · optional and ungraded — no separate deadline, nothing to submit; M1 is assembled from these five increments Keep in your group GitHub repo: stratum-specific estimates + a formal interaction test + a short interpretation.

### The question this week

*Does your main exposure–outcome association differ across a scientifically meaningful subgroup?* This week you take the main estimate you built in P2 — your current best estimate, before the design-aware re-estimate at P4 — and ask whether one number actually hides two, i.e. whether some group carries the effect.

### What to do

1. **Choose one effect modifier with a rationale.** Sex, age band, income, or a clinical subgroup your exposure cannot cause — and say *why* modification is biologically/socially plausible. One well-argued modifier beats a fishing expedition. The modifier must be something **your exposure cannot change**. The safe check is time: sex, age, genes, or a condition that began before the exposure. Being recorded at the same visit is not enough — ask whether your exposure could have caused it. A variable your exposure can change, such as a disease it causes, turns the question into mediation, not modification: stratifying on it removes part of the effect and compares unlike people (Week 5, “The modifier must survive the ‘could A change it?’ test”). Say which question you are asking: **effect modification** (your exposure’s effect, by M — adjust for your exposure’s confounders) or **interaction** (two exposures — adjust for the confounders of both).
2. **Count, then estimate stratum-specific effects.** Tabulate people and outcomes in every exposure  $\times$  modifier cell first; a sparse cell decides how much any interaction measure can tell you. Then fit the association within each level of the modifier and report the estimates side by side.
3. **Test for modification formally, on the right scale** (a product term is the tool).
  - *Multiplicative*: an exposure  $\times$  modifier product term in your outcome model.
  - *Additive* (often more relevant for public health): compare **absolute risks** across strata — for a time-to-event outcome, contrast predicted risks at a fixed horizon rather than forcing a logistic-model RERI.
4. **Interpret.** Report the stratum-specific estimates and the interaction test with their intervals, and state which scale you tested and why. A crude stratum difference is not an interaction test, and a null test is not evidence of homogeneity — say what the data show at the precision you have, and do not claim to have identified what produced any crude difference.

### Deliverable

- Stratum-specific estimates + the interaction test  $\rightarrow$  your M1 **effect-modification slide**.
- 2–3 sentences: is there modification, on which scale, and does it change the story?

### Worked example — MASLD/NHANES demonstration

- The paper reports **one** hazard ratio per phenotype and never tests modification. A natural P3 question: **does the central-adiposity  $\rightarrow$  mortality association differ by sex?** (Group II is 56% female; Group I only 15% — so sex is entangled with the exposure and worth probing.)
- **Right tool for the outcome:** mortality is time-to-event with censoring. Restrict to the locked contrast (Groups III and IV) and put an  $IV \times \text{sex}$  term in the **Cox** model, so the product term is a 1-df test of exactly that contrast (multiplicative). For the additive scale, compare **standardised 6-year mortality risks** of IV vs III within each sex — *not* a logistic RERI on an “ever died” flag.
- A defensible result is often “no meaningful modification” — reported clearly, that is a complete P3.

## What good looks like

- **Exceptional:** a modifier chosen for a stated scientific reason; correct stratum-specific estimates; interaction tested on the appropriate scale for the outcome type; reports the interval, not just the p-value, and does not claim to have identified what produced a crude stratum difference.
- **Adequate:** a sensible modifier assessed with a product term and strata.
- **Weak:** a product term with no rationale; or a logistic RERI that discards follow-up time and censoring.

## Be ready to defend (M3)

*“You report an interaction by sex. Is that genuine effect modification, or just what you’d expect from men and women having different baseline mortality? Defend the scale you tested it on.”*

## AI co-pilot note

Because the interaction *lab* uses a binary-outcome RERI example, a co-pilot will often copy that recipe — coding “ever died” and fitting logistic regression — which throws away survival time. For a time-to-event paper you must move the interaction into the survival model. Don’t let the lab’s convenient example dictate your project’s method.

## Lab → project handoff

You are applying **L4 (Interaction)** — product terms, stratum-specific estimates, and additive vs multiplicative modification (EpiMethods `confoundingE2`). Assess it on the scale your outcome demands.